Who bears the AI shock?
The distributional and fiscal incidence in the UK*

Vahid Ahmadi†

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Abstract

I estimate how much of an AI-driven labour-market shock the UK tax-benefit system absorbs and who bears the rest. I link occupation-level AI exposure to workers in Microcosm, a calibrated microdataset built from the 2024–25 Family Resources Survey, and use PolicyEngine UK to run displacement, survivor-wage and capital-income shocks through the tax-benefit system at household level. Two comparisons hold the size of the shock fixed. First, reallocating the same expected number of displaced workers (7 per cent of employees) across four stylised incidence rules changes the annual Exchequer cost from £2.2 billion to £11.2 billion; a top-loaded stress test reaches £26.9 billion. Second, delivering the same gross earnings loss through wage cuts rather than job losses raises the cost from £7.4 billion to £18.6 billion and lowers the rise in absolute poverty from 1.95 to 0.19 percentage points on one paired draw, whose poverty rise lies near the bottom of the 50-draw range: the Exchequer absorbs 42 pence per pound of earnings lost through job loss and 54 pence through wage cuts. Averaged over 50 draws, the exposure-proportional central case costs £7.4 billion a year, the net of a £40.7 billion displacement cost and receipts from assumed survivor-wage and capital-income gains; it turns into a net gain at a survivor-wage effect of 3.7 per cent.

Keywords: artificial intelligence, labour market, microsimulation, poverty, inequality, public finances

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†Research Associate at PolicyEngine. Email: vahid@policyengine.org.

1 Introduction

This paper estimates how much of an AI-driven labour-market shock the UK tax-benefit system absorbs and who bears the rest. The answer turns on two things the size of the shock does not settle: which workers lose earnings, and whether they lose their jobs or part of their pay. The occupations that score highest on AI exposure indices—professional, associate professional, administrative and managerial work—pay well and carry much of the UK's income tax and NI base (Felten et al., 2021; Eloundou et al., 2024; Department for Education, 2023; Henseke et al., 2025), while UC and other means-tested benefits respond mainly to losses lower down the distribution. The same aggregate shock can therefore cost the Exchequer a little or a lot, and raise poverty a little or a lot, depending on where it lands.

The evidence on where it will land is unsettled. Exposure-based scenarios place the risk on highly paid cognitive work (Felten et al., 2021; Eloundou et al., 2024); an expertise-compression view expects AI to erode the premia of experienced workers (Autor, 2024; Hosseini Maasoum and Lichtinger, 2026b); and early empirical studies find larger employment effects for junior workers in exposed occupations and firms (Klein Teeselink, 2025; Hosseini Maasoum and Lichtinger, 2026a; Brynjolfsson et al., 2026). The adjustment margin is equally open: in task-based models displacement can show up as relative wage declines rather than job loss (Acemoglu and Restrepo, 2022).

I run an AI labour-market shock through the UK tax-benefit system at household level and vary both dimensions while holding the shock's size fixed. Using PolicyEngine UK on Microcosm, a calibrated microdataset built from the Family Resources Survey 2024–25, I attach occupation-level exposure to workers and adopt the central conventions of an Irish study (Doorley et al., 2026): 7 per cent of employees displaced, a 2.6 per cent wage uplift for survivors and a higher return to capital. Two comparisons organise the results.

Who loses the job. Reallocating the same expected number of displaced workers across four stylised rules changes the annual Exchequer cost from £2.2 billion under uniform incidence to £11.2 billion when displacement tilts towards high earners in exposed occupations (50-draw means); a top-loaded stress test reaches £26.9 billion. Poverty does not follow cost: uniform incidence costs least but raises absolute poverty and the Gini by more than exposure-proportional incidence.

How the loss happens. Delivering the same gross earnings loss as occupation-level wage cuts rather than job losses raises the cost from £7.4 billion to £18.6 billion and lowers the rise in absolute poverty from 1.95 to 0.19 points (paired draw). The Exchequer absorbs 42 pence of each pound of earnings lost through job loss and about 54 pence of each pound lost through wage cuts.

The exposure-proportional central case is one cell of this design. It costs £7.4 billion a year as the net of a £40.7 billion displacement cost and extra receipts from the survivor-wage and capital channels, so its sign reverses with a survivor-wage effect of 3.7 per cent, a capital-income uplift of about 56 per cent or out-of-work spells of about 9.4 months. An explicit account of where the displaced wage bill goes, with corporation tax and an assumption about the UK-household ownership of profits, gives results from a net gain of £1.1 billion to a cost of £13.8 billion. The OBR's own technological-displacement scenario, in which equilibrium unemployment rises from 4.1 to 5.5 per cent, adds an average of £9 billion a year to borrowing (OBR, 2026a); displacing

employees on the same head-count scale, with flat earnings and no capital gain, costs £9.7 billion in the simulation year, almost all of it lost income tax and NI.

The paper contributes to three literatures. For the AI exposure literature it supplies the household tax-benefit layer that exposure indices and macroeconomic scenarios lack; I am not aware of a published study that runs an AI shock through the UK tax-benefit system at household level. For the literature on automatic stabilisers (Dolls et al., 2012) it compares the absorption of an extensive-margin and an intensive-margin shock of equal size. For the public finance of AI (Korinek and Lockwood, 2026; OBR, 2026b) it shows, household by household, how a shift from labour income to profits plays out under current UK rules. The code and aggregate results are public; exact regeneration requires access to Microcosm.

Section 2 positions the paper, Section 3 sets out the data and methods, Section 4 reports the results, Section 5 prices three policy responses and Section 6 concludes.

2 Related literature

Exposure and incidence. Occupation-level exposure measures link AI capabilities to the abilities or tasks of each occupation (Webb, 2019; Felten et al., 2021; Tolan et al., 2021; Eloundou et al., 2024; Gmyrek et al., 2023; ILO, 2025); UK applications include Department for Education (2023) and the task-based index of Henseke et al. (2025). The government's January 2026 assessment reports, from IMF estimates, that around 70 per cent of UK workers are in exposed jobs and about half of those in high-complementarity roles, and it gives no displacement estimate (DSIT and AISI, 2026). Exposure does not settle incidence. The complementarity adjustment of Pizzinelli et al. (2023) and Cazzaniga et al. (2024) separates exposure that substitutes for workers from exposure that augments them, and in a calibrated task model Rockall et al. (2025) find that AI can reduce wage inequality by displacing high-income workers while raising wealth inequality through higher capital returns. The expertise-compression view holds that AI lowers the premia of experienced workers by letting less experienced ones perform their tasks (Autor, 2024; Hosseini Maasoum and Lichtinger, 2026b), consistent with experimental gains concentrated among less experienced workers (Noy and Zhang, 2023; Brynjolfsson et al., 2025). Early UK and US evidence points instead to junior workers: Klein Teeselink (2025) reports larger employment falls for junior positions at more exposed UK firms, Hosseini Maasoum and Lichtinger (2026a) find junior employment falling after firm adoption in US résumé data, and in their latest version Brynjolfsson et al. (2026) report that employment of 22–25-year-olds in the most exposed US jobs stands 19 per cent below the path of less exposed peers, a pattern they describe as descriptive rather than causal. Danish administrative data rule out effects on earnings or recorded hours larger than 2 per cent two years after ChatGPT's launch (Humlum and Vestergaard, 2025), US vacancy data show exposed establishments changing their hiring without detectable aggregate employment effects (Acemoglu et al., 2022), and task-based estimates imply modest macroeconomic gains over a decade (Acemoglu, 2025). Because these views imply different incidence, I vary incidence rather than assume it.

Labour-market shocks in tax-benefit models. Dolls et al. (2012) define the income stabilisation coefficient, the share of a market-income shock absorbed by taxes and transfers, and find that automatic stabilisers absorb 38 per cent of a proportional income shock in the EU and 32 per cent in the US, against 47 and 34 per cent of an unemployment shock. Brewer and Tasseva (2021) run the Covid-19 employment shock and policy response through UKMOD. Doorley

et al. (2023) find that European tax-benefit systems muted the pass-through of routine-biased automation to household inequality, and Doorley et al. (2026) run the 7 per cent displacement convention through the Irish SWITCH model, comparing exposure-proportional and uniform allocation. Evidence on the persistence of UK earnings losses after job loss (Upward and Wright, 2019) bears on the duration assumption. I add the UK, a fixed-quota comparison of five allocation rules with paired draws, and a comparison of job loss with wage cuts at the same gross earnings loss, which is the extensive-versus-intensive contrast of Dolls et al. (2012) applied to an AI shock.

The public finance of AI. Task-based models imply that automation lowers the labour share (Acemoglu and Restrepo, 2018) and raises returns to wealth (Moll et al., 2022); if AI shifts income from labour to capital, labour-tax bases erode (Korinek and Lockwood, 2026). The OBR models this in a technological-displacement scenario in its March 2026 outlook (OBR, 2026a) and in a long-term scenario in which the wage share halves (OBR, 2026b); one US exercise runs alternative AI futures through the federal tax system without benefits (Iselin and Nunn, 2026). Brollo et al. (2024) discuss unemployment insurance and capital taxation as fiscal responses to generative AI. A normative literature asks whether automation should be taxed (Guerreiro et al., 2022; Costinot and Werning, 2023; Thuemmel, 2023), documents that the US tax code taxes labour more heavily than equipment (Acemoglu et al., 2020), and shows that slow reallocation and borrowing constraints can make automation inefficient (Beraja and Zorzi, 2025); Bastani and Waldenström (2024) review capital taxation in this setting. This paper is positive rather than normative. It supplies the household tax-benefit layer under the OBR's aggregate scenarios: who loses what, how much the Exchequer absorbs, and how that depends on incidence and margin.

3 Data and methods

The analysis has three stages. I attach an occupation-level measure of AI exposure to workers in a calibrated UK household microdataset, translate an assumed economy-wide shock into person-level changes to employment, wages and capital income, and pass baseline and shocked data through a tax-benefit microsimulation model.

3.1 Measuring AI exposure

I start from the AI Occupational Exposure (AIOE) index of Felten et al. (2021), which links progress in ten AI application areas to the ability requirements of each occupation in O*NET. The AIOE is standardised across occupations (mean zero, unit standard deviation), so it takes negative values for occupations with below-average exposure. I combine it with the complementarity index $\theta_l \in [0, 1]$ of Pizzinelli et al. (2023), built from O*NET work-context descriptors and job-zone requirements; high-θ occupations are those in which AI is more likely to complement than replace human judgement. The complementarity-adjusted exposure of occupation group l is

$$\text{C-AIOE}_l = \text{AIOE}_l \times \big(1 - (\theta_l - \theta_{\min})\big), \tag{1}$$

where $\theta_{\min}$ is the lowest complementarity among the groups. Because the standardised AIOE can be negative, so can the C-AIOE (Table 1). Before allocating job losses I therefore shift the

scores so that the least-exposed group scores zero, $\widetilde{\mathrm{C}}_l = \text{C-AIOE}_l - \min_k \text{C-AIOE}_k \geq 0$, and the allocation rule uses $\widetilde{\mathrm{C}}_l$.

No published UK C-AIOE series exists at the level I need, so I build one. AIOE scores come from the UK SOC2020 mapping published with Department for Education (2023), which chains the Felten et al. (2021) scores through US SOC2018, US SOC2010 and ISCO-08 to UK SOC2020. I reconstruct θ from the public O*NET 27.3 release following Pizzinelli et al. (2023); the reconstruction reproduces the published ordering up to a level offset, which does not matter because equations (1) and (3) use θ only relative to its minimum and its mean. Both series are aggregated to the nine SOC2020 major groups with ASHE 2025 Table 14 employment weights, which cover 340 of the 412 unit groups. Exposure therefore varies across nine groups only, and all within-group variation is lost; the dataset's sub-major group is imputed, and I do not use it.

3.2 Shock scenarios

The size of an AI shock cannot be estimated from historical data, so I treat it as a scenario input. The central calibration follows the Irish design of Doorley et al. (2026): 7 per cent of employees lose their jobs and surviving employees receive an average wage uplift of 2.6 per cent. Doorley et al. derive these inputs from one investment-bank scenario exercise (Briggs and Kodnani, 2023), which describes displacement over a roughly decade-long adoption period with reallocation. Applying the whole 7 per cent within one simulated year turns that flow into a stock, so the central case is an impact-year stress test, not an annual forecast. For scale, the OBR estimates that around 10 per cent of the UK labour force could be exposed to substitution from AI over the next ten years, and a further 30 per cent could have their work complemented (OBR, 2026b, Box 4.1). I also follow Doorley et al. (2026) in raising the return to capital by 0.4 percentage points, from 1.005 to 1.405 per cent, which they motivate with Cazzaniga et al. (2024). Around the central case, a grid crosses displacement rates of 0–10 per cent with wage uplifts of 0–5 per cent (66 cells, capital shock on throughout; Online Appendix B.5), and two presets add a 1 per cent sensitivity case with no wage gain and a 13 per cent stress case.

3.3 Implementing the shock at the micro level

Employment. The number displaced is the displacement rate times weighted employees. It is allocated across groups in proportion to employment times shifted exposure:

$$\text{JobLoss}_l = \text{TotalJobLoss} \times \frac{\text{EMP}_l \, \widetilde{\mathrm{C}}_l}{\sum_k \text{EMP}_k \, \widetilde{\mathrm{C}}_k}. \tag{2}$$

Table 1 reports the resulting quota rate for each group. The rule concentrates losses in administrative and secretarial, managerial, professional and associate professional occupations, and elementary occupations, the least-exposed group, receive no displacement. Employees without a recorded group form a pseudo-group with the employment-weighted mean exposure. Within each group, systematic sampling on a random permutation selects the displaced: every record in a group has the same inclusion probability (up to prescribed tilts such as the youth multiplier), the quota holds in expectation, and a record's calibrated weight enters only through the quota accounting. A displaced worker's employment income, hours, pension contributions (including salary sacrifice) and statutory pay are set to zero and their status to unemployed; dual jobholders keep self-employment income. PolicyEngine UK takes no spell-duration input, so displaced

workers are assessed as unemployed for the whole year. The self-employed are outside both labour-market shocks, following Doorley et al. (2026).

Table 1: Central-scenario job loss by SOC2020 major group

Major group	C-AIOE	Employment (m)	Quota rate (%)
1 Managers, directors and senior officials	0.62	3.33	9.4
2 Professional occupations	0.60	8.12	9.3
3 Associate professional occupations	0.47	4.25	8.4
4 Administrative and secretarial occupations	0.74	2.98	10.3
5 Skilled trades occupations	−0.33	1.89	2.8
6 Caring, leisure and other service occupations	−0.14	2.42	4.1
7 Sales and customer service occupations	0.27	1.79	7.0
8 Process, plant and machine operatives	−0.54	1.42	1.3
9 Elementary occupations	−0.73	2.69	0.0

Note: C-AIOE is the standardised complementarity-adjusted exposure score before the min-zero shift used in the allocation rule. Employment uses calibrated weights. Quota rates are the prescribed group displacement rates; a single draw realises them in expectation only.

Wages. Each surviving employee receives an uplift proportional to complementarity:

$$\text{WageChange}_l = \text{WageGrowth} \times \frac{\theta_l}{\bar{\theta}} \times \text{Wage}_l, \qquad \bar{\theta} = \frac{\sum_l \text{EMP}_l\, \theta_l}{\sum_l \text{EMP}_l}, \tag{3}$$

with EMP_l baseline employment. The employment-weighted mean uplift equals the assumed growth rate. Two features follow from the rule. Higher-θ occupations earn more, so the wage bill grows slightly faster than the headline rate, and survivors have above-average θ, so displacement raises the realised mean uplift a little. Non-displaced workers can only gain; Online Appendix B.6 prices displaced workers bidding down others' wages in reduced form.

Capital. Savings interest and dividend income are scaled by 1.405/1.005, a 39.8 per cent increase; rental income and returns inside pensions are unchanged. The baseline return of 1.005 per cent comes from the Irish calibration. The same 0.4-point shock on a 4 per cent base would raise capital incomes by 10 per cent, so the channel is best read directly as a 39.8 per cent proportional increase. Section 4.6 replaces it with an explicit account of where the displaced wage bill goes.

3.4 Incidence and adjustment-margin scenarios

Holding the central aggregate shock fixed, I reallocate the same expected number of displaced workers under five displacement rules. Survivor wages and capital income are held fixed at the person level, so the comparisons isolate who loses the job.

Exposure-proportional. Equation (2); the central case and the rule behind the grid.

Junior-concentrated. Draws tilted towards workers aged 16–24 by a multiplier of 1.29, the ratio of junior to overall employment falls in press-release figures for maximally exposed UK firms from Klein Teeselink (2025) (5.8/4.5), within unchanged group quotas.

Expertise-compression. Displacement probabilities doubled (an author choice) for the top weighted-earnings tertile of workers in above-median-exposure occupations: a high-earner displacement tilt named after the hypothesis that motivates it (Autor, 2024; Hosseini Maasoum and Lichtinger, 2026b), whose formal mechanisms work through wages rather than job loss.

Uniform. Every employee has the same displacement probability.

Top-loaded stress test. Probability proportional to shifted exposure times a junior factor of 1.29 and wage-tier factors of 9.6 above and 0.6 below weighted-median person earnings. The tier factors carry press-release figures for maximally exposed UK firms (employment falls of 9.6 per cent in high-paying firms and almost none in low-paying ones) over to persons, using person earnings as a proxy for firm pay (Klein Teeselink, 2025). I treat this family as an upper stress case, not an estimated schedule.

Wage margin. A separate family moves the same earnings loss to wage cuts (Acemoglu and Restrepo, 2022). No one loses a job; every employee's earnings fall by an occupation-level percentage proportional to $\widetilde{C}_l$, scaled so that the aggregate gross loss equals the realised loss of the paired central displacement draw (7.71 per cent of the baseline employee wage bill at seed 0, above 7 per cent because group job-loss rates rise with group pay). The 2.6 per cent uplift and the capital shock apply on top. A second gradient uses the potential-supply-shift measure of Hosseini Maasoum and Lichtinger (2026b) to rank the cuts. A mixed family thins the central displacement draw to a share $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$ and fills the gap to the same gross loss with C-AIOE-graded cuts, and a survivor-wage panel holds the central draw fixed while the survivor-wage effect runs from -5 to $+5$ per cent.

3.5 Microsimulation with PolicyEngine UK

I use PolicyEngine UK 2.174.1 (policyengine.py 6.16.1 release bundle) on Microcosm UK 2024–25 (build `microcosm-uk-2024-25-national-20261107T020434Z-eeb6093e`). Microcosm starts from the Family Resources Survey 2024–25 (UK Data Service SN 9563), adds records drawn from the HMRC Survey of Personal Incomes (SPI) for top incomes, imputes variables the survey does not collect, and calibrates household weights to 1,514 administrative and survey targets, including SPI income by band up to £1 million, ONS occupational employment, DWP caseloads and OBR receipts; 98.7 per cent of targets fall within 10 per cent.[1] The file holds 78,181 households and 182,418 persons, grossing to 69.3 million people and 29.0 million employees in 2026; 99.7 per cent of weighted employees have a recorded SOC2020 major group, carried from the FRS and imputed from income for SPI-drawn records. The simulation year is 2026.

Each scenario runs a baseline and a shocked simulation on identical records, so reported effects contain no resampling noise, but they inherit the survey's sampling error. Income results use HBAI cash disposable income. Absolute poverty lines are fixed at baseline values; relative lines are 60 per cent of the shocked median. The Gini is computed on equivalised household income with negative incomes set to zero. The Exchequer cost is the fall in PolicyEngine's government balance, which includes income tax, employee and employer National Insurance (NI) and benefit spending; a negative cost is a net gain. Household consumption is held at baseline, so the balance carries no VAT or other indirect-tax response, and the pay bill saved

[1] Target counts and fit are from the build's calibration diagnostics in the Microcosm repository.

when a public-sector employee is displaced is outside the balance; I report that pay bill (earnings, employer NI and employer pension contributions of displaced public-sector employees, identified by the FRS employment sector) beside the cost. The household basis counts only the items between household market income and disposable income that fall on the household: income tax, employee NI and benefits, which excludes employer NI. Its share of a market-income change is the income stabilisation coefficient of Dolls et al. (2012).

Draws and uncertainty. The identity of the displaced is random. Calibration makes Microcosm's weights uneven: the 72,504 employee records have a Kish effective sample size of about 5,810, so the realised number displaced varies around the quota (1.85–2.24 million over 50 central draws). I report 50-draw means for the central case, the incidence families and the policy reforms; 20-draw means for the channel split, the capital-income coverage and capital variants, the public-sector pay bill, the OBR benchmark and the duration and take-up sensitivities; and single draws at seed 0 where only one draw exists (the grid, the presets other than the central case, the wage-margin and mixed tables, and the index and ripple sensitivities), labelled as such. A value written "$a \pm b$" is a mean and the standard deviation across assignment draws, which describes how much a single draw can vary; it is not a confidence interval. The Monte Carlo standard error of a 50-draw mean is that standard deviation divided by $\sqrt{50}$: £0.26 billion for the central cost. Neither number reflects survey sampling or parameter uncertainty. Appendix A.1 lists the basis of every result.

4 Results

I first report the central scenario, how the tax-benefit system absorbs it and where in the income distribution the loss falls, then the two comparisons that organise the paper: who loses the job, and how the earnings loss arrives. The section closes with the assumptions that move the central cost, an account of where the displaced wage bill goes, and a benchmark against the OBR.

4.1 The central scenario and its three channels

Table 2 summarises the central scenario and the two presets. Over 50 assignment draws, the central shock displaces 2.04 million employees, costs the Exchequer £7.4 billion a year (draw standard deviation £1.8 billion; standard error of the mean £0.26 billion), and raises the Gini coefficient by 1.76 points from a baseline of 0.302. The exposure-proportional rule concentrates job loss in higher-income households: 0.80 per cent of people in the bottom decile live through a transition to unemployment, against 5.55 per cent in the top (Figure 1). That gradient restates the allocation assumption; it is not evidence about who AI will displace.

The central cost nets three large flows. Running each channel alone on 20 paired draws, displacement costs £40.7 billion, while the survivor-wage and capital channels each raise net receipts by £16.8 billion; the channels sum to within £0.15 billion of the full shock. Per displaced worker, the displacement channel costs £20,200 (mean of the per-draw ratio over 20 draws, standard deviation £1,000). By tax and benefit, the 50-draw cost combines lost income tax on earnings, net of the tax on survivors' wage gains (£10.5 billion), employer NI (£8.6 billion), employee NI (£2.7 billion) and higher Universal Credit (UC) spending (£2.3 billion), against higher dividend tax (£12.7 billion), higher savings tax (£3.0 billion) and lower childcare support (£1.0 billion). The Exchequer loses the employer contributions on displaced workers' earnings

Table 2: Preset scenarios

Scenario	Displaced (million)	Exchequer cost (£bn/yr)	Δ Poverty (pp) Abs. BHC	Abs. AHC	Rel. BHC	Δ Gini (pp)	Public pay bill
Low (seed 0)	0.29	−10.4	−0.01	+0.11	–	+0.73	–
Central (50-draw mean)	2.04	7.4	+2.34	+2.58	+0.64	+1.76	25.8
High (seed 0)	3.85	43.4	+5.29	+5.64	–	+2.81	–

Note: Exchequer cost is the fall in the government balance; a negative cost is a net gain. Low: 1 per cent displacement, no wage gain. Central: 7 per cent displacement, 2.6 per cent wage gain. High: 13 per cent displacement, 2.6 per cent wage gain. The capital shock applies in every row. Abs.: absolute line fixed at baseline; Rel.: 60 per cent of the shocked median. The relative line is reported for the central case only. Public pay bill: earnings, employer NI and employer pension contributions of displaced public-sector employees (£bn, mean over 20 draws), reported beside the cost and not netted from it.

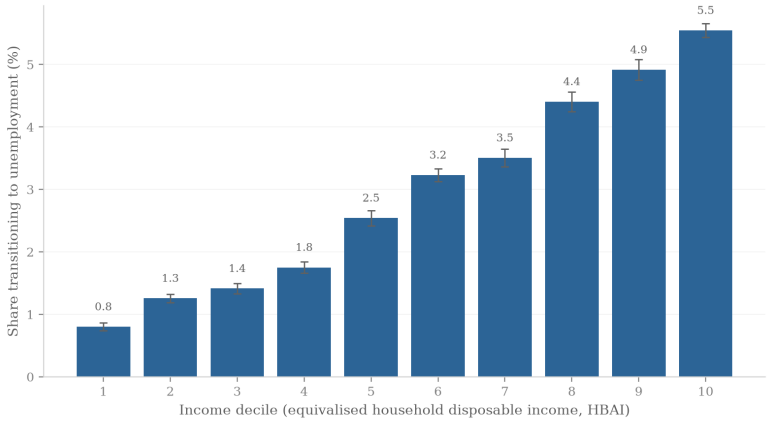


Figure 1: Share of each equivalised disposable income decile that moves from employment to unemployment, central scenario. Means over 50 draws with 95 per cent intervals for the mean.

whatever their incidence; whether the saved labour cost returns as taxable profit is the closure question of Section 4.6. The government balance also omits the public sector's side of displacement: displaced public-sector employees cost £25.8 billion a year in pay, employer NI and employer pension contributions (mean over 20 draws, standard deviation £2.1 billion; Table 2). Whether that sum is a saving depends on whether the services they provided continue, which the simulation does not model.

The Gini rises because displaced workers lose all earnings, survivors gain, and the capital uplift accrues mainly in the top decile, which holds 61.2 per cent of baseline capital income. The rise does not depend on the capital channel: on its own the displacement channel raises the Gini by 0.86 points (20 draws), and with the capital shock switched off the Gini rises in all 65 shocked cells of the grid (Online Appendix B.5).

Poverty. Absolute BHC poverty rises by 2.34 points (standard error 0.03). I read this as an upper bound. Every displaced worker loses a full year of earnings against a poverty line frozen at its baseline value. On the relative line, which moves with the shocked median, the central rise is 0.64 points.

A historical benchmark. The framework over-predicts the poverty response to a past downturn. A backcast displaces 2.6 per cent of employees uniformly, matching the rise in the LFS unemployment rate over 2008–2010, with no wage or capital channel (means over five draws). It raises simulated absolute BHC poverty by +0.97 points, absolute AHC poverty by +1.17 points and the Gini by +0.33 points, and changes relative BHC poverty by −0.19 points. Recorded HBAI outturns over 2007/08–2010/11 show absolute BHC poverty and the Gini roughly flat to slightly falling and relative poverty falling: benefits were uprated with high inflation while median incomes fell, and unemployment spells were of mixed length. The gap measures what the channels outside the simulation (uprating, duration, behavioural and macro-policy responses) absorbed in practice. Per point of uniform displacement the backcast gives +0.37 points of BHC poverty, against +0.35 for the uniform 7 per cent family, which also carries the wage and capital channels.

What a displaced worker can claim. A displaced employee meets the UK system through two routes. New-style Jobseeker's Allowance, a contributory benefit, pays £95.55 a week to a claimant aged 25 or over for up to 26 weeks; the simulation pays it only where the survey reports an award, so newly displaced workers do not receive it (Section 6.1). Universal Credit (UC) is means-tested on the benefit unit. In 2026–27 its standard allowance is £424.90 a month for a single claimant aged 25 or over and £666.97 for a couple with one partner aged 25 or over, plus elements for children, housing, disability and childcare. A partner's earnings above the work allowance (£427 a month with housing support and £710 without, for claimants with children or limited capability for work) reduce the award by 55 pence in the pound, and a benefit unit with capital above £16,000 has no entitlement. Each benefit unit keeps the UC take-up decision drawn for it in the baseline data. A displaced sole earner with no other income and capital below the limit is entitled to the full standard allowance and, if renting, the housing element; a partner's earnings and the household's capital enter the same test.

Caseloads. In the seed-0 central draw the shock draws 351,000 benefit units into UC while 18,000 exit, a net increase of 333,000, against 1.96 million displaced workers. Annual UC spend-

ing rises by £2.4 billion from a baseline of £80.5 billion, of which £0.6 billion is the housing element within paid awards. The high preset adds 506,000 net benefit units and £4.1 billion, the low preset 87,000 and £0.8 billion, and the junior-concentrated rule 293,000 and £2.3 billion. Only the benefit-unit figures are net of exits; the corresponding counts of households (314,000) and persons (664,000) are gross new entrants.

4.2 Where the income loss falls

Figure 2 reports the central-scenario change in disposable income by baseline decile, averaged over 20 displacement draws, with each household weighted by its number of members. The profile is hump-shaped: the largest mean loss, 2.86 per cent, falls in decile 8, because displacement losses concentrate in the upper-middle deciles while capital-income gains offset them at the top, where the change is +0.09 per cent (95 per cent interval for the mean $[-0.14, +0.32]$). The bottom-decile result depends on how households are ranked. Under the HBAI ranking decile 1 changes by +1.78 per cent; ranked on income before pension contributions it changes by −0.33 per cent, and excluding the 333 households whose personal pension contributions exceed both £10,000 and their earnings it changes by −0.10 per cent. A small number of heavily weighted households with large personal pension contributions sit in decile 1 under HBAI and receive capital income, so the bottom-decile change is not robust to how households are ranked, and I do not report it as a finding.

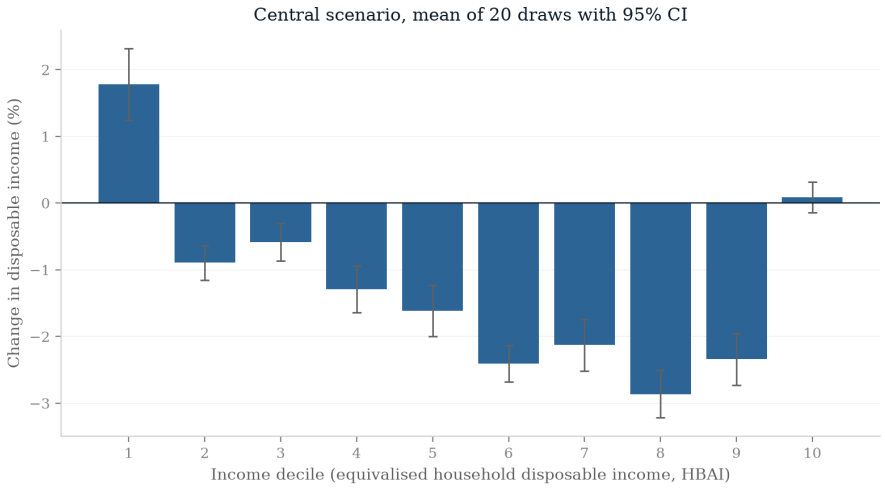


Figure 2: Change in household disposable income by baseline decile, central scenario: mean and 95 per cent interval for the mean across 20 assignment draws. Each household's change is weighted by its number of members.

Figure 3 splits the change into market income, benefits, and taxes and contributions, all relative to baseline disposable income, on one draw (seed 0) that weights households once. Market income change plus benefit change minus tax change equals the disposable-income change up to a residual, which collects perimeter items inside the HBAI concept (council tax and pension deductions) and never exceeds 0.5 per cent of baseline income in any decile. In decile 8, which loses most in this draw, market income changes by −4.9 per cent of baseline disposable income, benefits by +0.3 per cent and lower taxes and contributions return +0.7 per cent, leaving a

disposable-income change of −3.6 per cent. Benefits and lower taxes offset about 21 per cent of the market loss there, and 27 per cent including the perimeter items. In the top decile the capital uplift offsets the earnings lost by displaced workers: market income changes by +0.8 per cent and disposable income by +0.4 per cent.

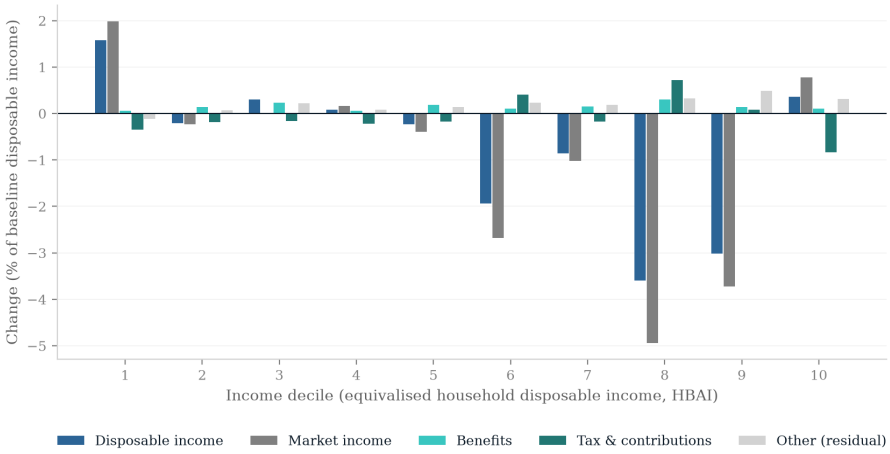


Figure 3: Change in market income, benefits, and taxes and contributions by baseline decile, central scenario (single draw, seed 0), as a share of baseline HBAI disposable income. The residual collects perimeter items (council tax, pension deductions) inside the HBAI concept.

The central scenario mixes displacement with survivor-wage and capital gains, which blurs how much of a job loss the system absorbs. Table 3 isolates job loss. It uses the OBR benchmark run of Section 4.7, which displaces 1.67 per cent of employees under exposure-proportional incidence with flat survivor earnings and no capital shock, and sums household flows by baseline decile over its 20 draws. On the household basis, income tax, employee NI and benefits absorb 28 per cent of the £23.3 billion market-income loss, and 34 per cent of it does not reach HBAI disposable income once pension contributions and the other perimeter items count. The tax-benefit share runs from 19 to 25 per cent across deciles 2–9 and reaches 37 per cent in the top decile, which carries 37 per cent of the market loss and pays higher-rate income tax on it. Taxes carry almost all of the absorption: UC rises by £0.67 billion, 2.9 per cent of the earnings lost.

On the household basis, income tax, employee NI and benefits absorb 28 per cent of the market-income loss from job loss here; Dolls et al. (2012) report the corresponding shares for an unemployment shock in the EU and the US (Section 2). Two differences in what the shock triggers bear on the comparison. Their unemployment shock pays unemployment insurance where entitlement exists, whereas the simulation pays no new-style JSA to newly displaced workers. And exposure-proportional incidence places most of the market-income loss in the upper half of the distribution (Table 3).

4.3 Who loses the job

Table 4 reallocates the same expected number of displaced workers across the five rules of Section 3.4; survivor wages and capital income are identical at the person level. Across the four stylised rules the Exchequer cost runs from £2.2 billion (uniform) to £11.2 billion (expertise compression); the top-loaded stress test costs £26.9 billion. The cost tracks the earnings of

Table 3: Household absorption of job loss by baseline decile (OBR-sized displacement, 20 draws)

	Change (£bn a year)				Absorbed (%)	
Decile	Market income	Benefits	Income tax and NI	Disposable income	Tax and benefits	$1 - \Delta Y_D / \Delta Y_M$
1	−0.35	0.01	−0.08	−0.09	26	74
2	−0.44	0.03	−0.06	−0.31	19	28
3	−0.77	0.06	−0.13	−0.53	25	31
4	−0.85	0.04	−0.14	−0.61	22	28
5	−1.16	0.08	−0.19	−0.82	23	29
6	−1.63	0.07	−0.30	−1.16	23	29
7	−2.47	0.07	−0.50	−1.78	23	28
8	−3.00	0.03	−0.65	−2.15	23	28
9	−3.97	0.00	−0.99	−2.75	25	31
10	−8.67	0.03	−3.15	−5.18	37	40
All	−23.31	0.42	−6.20	−15.38	28	34

Note: Means over 20 draws of household flows summed within baseline deciles of equivalised HBAI income (households ranked with person weights, summed with household weights). Income tax and NI are household income tax and employee NI. "Tax and benefits" is (change in benefits − change in income tax and NI) divided by the market-income loss; the last column also counts the other items between market and HBAI disposable income, chiefly pension contributions, which job loss removes. Decile 1 is subject to the ranking caveat above.

those displaced. Adding the survivor-wage and capital channels back, the displacement channel costs about £17,600 per displaced worker under uniform incidence, £20,200 under exposure-proportional, £22,000 under expertise compression and £29,800 under the stress test (50-draw means, assuming the channels add up, as they do to within £0.15 billion in the central case).

Table 4: The central shock under five displacement rules (50 paired draws)

Displacement rule	Displaced (million)	Exchequer cost (£bn/yr)	Δ Poverty (pp) BHC	AHC	Δ Gini (pp)
Exposure-proportional	2.04	7.4 ± 1.8	+2.34 ± 0.21	+2.58 ± 0.21	+1.76 ± 0.12
Junior-concentrated	2.03	7.1 ± 1.6	+2.37 ± 0.23	+2.61 ± 0.25	+1.77 ± 0.11
Expertise-compression	2.04	11.2 ± 2.5	+2.60 ± 0.20	+2.81 ± 0.23	+1.78 ± 0.11
Uniform	2.04	2.2 ± 1.9	+2.45 ± 0.22	+2.60 ± 0.22	+1.88 ± 0.12
Top-loaded stress test	2.03	26.9 ± 2.6	+2.93 ± 0.26	+3.39 ± 0.29	+1.77 ± 0.14

Note: Means ± standard deviations across 50 shared assignment draws; divide a standard deviation by $\sqrt{50}$ for the standard error of the mean. Every rule has the same expected displaced count. The top-loaded stress test multiplies exposure-proportional probabilities by a junior factor (1.29) and wage-tier factors (9.6 above and 0.6 below median person earnings).

Because the rules share assignment seeds, I compare them draw by draw. Uniform incidence costs £5.17 billion less than exposure-proportional incidence (paired 95 per cent interval $[-5.93, -4.41]$) and expertise compression £3.80 billion more ($[+2.92, +4.67]$); the junior tilt changes the cost by −0.29 billion ($[-0.97, +0.40]$), because the tilt operates within fixed occupation quotas and young workers sit mostly in low-exposure groups. These intervals describe assignment noise across 50 draws and narrow as draws are added; they carry no information about survey sampling or about which rule is right.

Poverty does not follow cost. Relative to exposure-proportional incidence, uniform incidence raises BHC poverty by a further +0.11 points ([+0.04, +0.19]) and the Gini by +0.12 points while costing less, and expertise compression raises poverty by +0.26 points more ([+0.19, +0.33]) while costing more. Two routes could produce this ordering. Uniform incidence displaces more workers in lower-income households, which start closer to the line; the high-earner tilts displace more sole earners, whose households fall the full distance. The saved draws do not separate the two routes.

Figure 4 shows the decile pattern behind these differences. Under every rule the share of each decile moving to unemployment rises with income, because employees concentrate in the higher deciles. Uniform incidence moves a larger share of the lower and middle deciles to unemployment than exposure-proportional incidence, and the top-loaded stress test a larger share of the top decile. The right panel places each rule's cost against its poverty change on the same single draw.

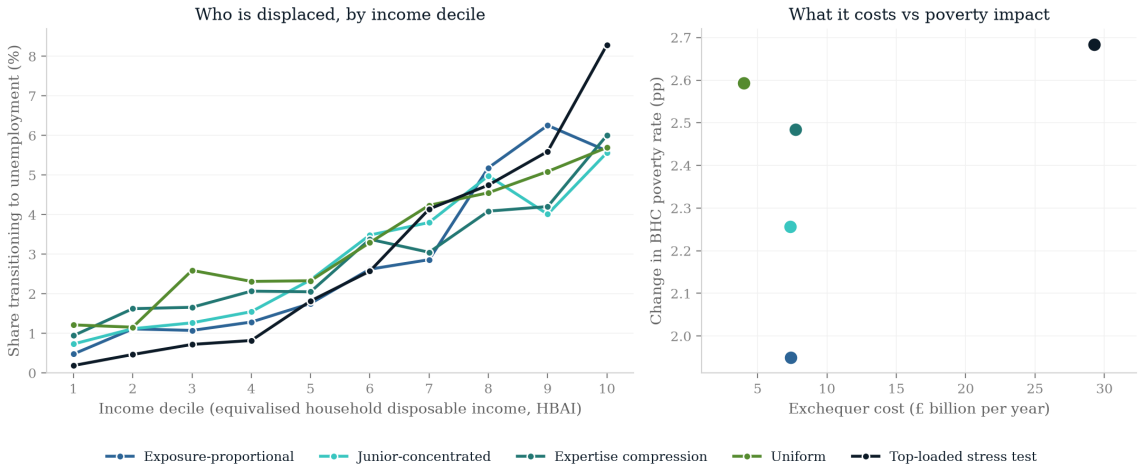


Figure 4: The central shock under the five displacement rules: share of each baseline decile moving to unemployment (left) and Exchequer cost against the change in absolute BHC poverty (right). Single draw (seed 0); Table 4 reports 50-draw means.

Age and gender. Young workers are concentrated in low-exposure occupations: 62 per cent of employees aged 16–24 work in the lowest exposure tertile, which is why the junior tilt moves the cost little. If AI reaches juniors mainly through reduced hiring (Hosseini Maasoum and Lichtinger, 2026a; Brynjolfsson et al., 2026), the stock-based framework does not capture it. Exposure also carries a gender gradient. Employed women have a mean C-AIOE of 0.33 against 0.22 for employed men, reflecting women's concentration in administrative, professional and associate professional work. Over 20 central draws the displacement rate is 7.6 per cent for employed women and 6.4 per cent for men, so women, who hold 49.1 per cent of employment, account for 53.4 per cent of the displaced, in line with international evidence that female employment is more exposed to generative AI (ILO, 2025).

4.4 How the earnings loss arrives

Table 5 moves the same gross earnings loss from job loss to wage cuts. The comparison is paired at seed 0, whose BHC poverty rise ranks 2 of 50 among the exposure-proportional draws, against a 50-draw mean of 2.34 points; seed 0 is therefore a low draw for poverty and a typical one for cost. Its displacement draw removes 7.71 per cent of the baseline wage bill and costs £7.4 billion (close to the 50-draw mean). Delivered as occupation-level wage cuts, the same loss costs £18.6 billion and raises BHC poverty by only +0.19 points, against +1.95 under job loss.

Table 5: The same gross earnings loss on two margins (paired at seed 0)

Scenario	Exchequer cost (£bn/yr)	Δ Poverty (pp) BHC	AHC	Δ Gini (pp)
Central displacement	7.4	+1.95	+2.15	+1.50
Wage margin (C-AIOE gradient)	18.6	+0.19	+0.34	+0.29
Wage margin (PSS gradient)	18.2	+0.18	+0.53	+0.34

Note: Every row removes 7.71 per cent of the baseline employee wage bill before the 2.6 per cent uplift and the capital shock, which apply in all rows. The wage-margin rows displace no one; cuts are proportional to the stated occupation gradient.

The margin changes how much of each lost pound the Exchequer absorbs. Displacement alone (seed 0, no wage or capital channel) removes £97.1 billion of earnings and costs the government balance 42 pence per pound; on the household basis of Section 3.5 it costs 29 pence. Cutting survivors' wages from the survivor-wage grid's 0 to −2.6 per cent row costs 54 pence per pound on the government balance; the saved draws do not split this figure into household and employer components. A wage cut removes pounds taxed at the worker's marginal rates of income tax and NI, and withdraws UC at the taper; job loss removes whole earnings, including the slice covered by the personal allowance and NI thresholds, and the UC it adds is small relative to the tax it forgoes. Poverty moves the other way: concentrated losses push displaced households across the line, while the same loss spread across 29.0 million employees moves few households below it.

Table 6 holds the gross loss fixed and moves it between margins. As the displaced share λ rises from 0 to 1, the cost falls from £18.6 billion to £7.4 billion and BHC poverty rises from +0.19 to +1.95 points. The second panel shows the survivor-wage assumption: with no pass-through to survivors the central cost is £24.2 billion, and a 2.6 per cent survivor-wage decline gives £41.0 billion.

4.5 What moves the central cost

The central cost is small relative to its components, so modest changes in three assumptions remove it. Table 7 reports the value of each at which the central cost is zero, holding the other two at their central values.

Other assumptions move the cost less or in one direction. A flat 70 per cent UC take-up probability, against Microcosm's person-weighted baseline claim propensity of 46 per cent, raises the cost from £7.2 billion to £8.3 billion (20 draws), and a reduced-form ripple effect in which displaced workers lower other workers' wages with an assumed slope of 0.3 raises it to £19.5 billion (seed 0; Online Appendix B.6). The low and high presets of Table 2 keep the full capital uplift while scaling displacement, which is why the low preset shows a net gain.

Table 6: Adjustment-margin and survivor-wage sensitivities (seed 0)

Scenario	Displaced (million)	Exchequer cost (£bn/yr)	Δ Poverty BHC (pp)	Δ Gini (pp)
Fixed gross earnings loss (7.71 per cent); λ is the displaced share				
$\lambda = 0$	0.00	18.6	+0.19	+0.29
$\lambda = .25$	0.52	15.4	+0.60	+0.60
$\lambda = .50$	0.93	13.3	+0.99	+0.82
$\lambda = .75$	1.42	10.5	+1.38	+1.10
$\lambda = 1$	1.96	7.4	+1.95	+1.50
Central displacement draw with alternative survivor-wage effects				
−5.0%	1.96	56.5	+2.75	+1.12
−2.6%	1.96	41.0	+2.48	+1.23
0.0%	1.96	24.2	+2.14	+1.36
+2.6%	1.96	7.4	+1.95	+1.50
+5.0%	1.96	−8.3	+1.64	+1.63

Note: In the first panel C-AIOE-graded survivor cuts fill the gap between the thinned displacement draw and the fixed gross loss; $\lambda = 1$ is the central draw. The second panel holds the displaced and the capital shock fixed and changes the mean survivor-wage effect, keeping the complementarity gradient.

Table 7: Break-even values of the central net cost

Assumption	Central	Break-even	Basis and method
Mean survivor-wage effect	+2.6%	+3.7%	Seed 0; linear interpolation between the +2.6 and +5 per cent rows of Table 6
Capital-income uplift	+39.8% (+0.40pp)	+56% (+0.57pp)	20 paired draws; linear between the central run and a simulated +60 per cent uplift of interest and dividends
Months out of work	12	9.4	20 draws; linear between full-year loss and the 50 per cent earnings-retention run, read as six months

Note: The capital figure interpolates between costs of £7.2 billion at +39.8 per cent and −1.6 billion at +60 per cent (means over the same 20 seeds). The duration figure rests on a run that halves annual earnings while keeping full-year unemployed status, so annual earnings withdraw UC that a real six-month spell would pay in its out-of-work months; it is approximate. Sources: results/robustness/survivor_wage_grid.csv, dividend_coverage_draws.csv, duration_takeup_sensitivity.json; results/next_runs/capital_variants_draws.csv.

Dividend coverage. Because the capital channel scales dividends proportionally, its fiscal size depends on how much dividend income the data carry. Restricted to people with total income of at least £12,570, the population the HMRC Survey of Personal Incomes covers, Microcosm's 2026 dividend income is £92.7 billion, 16 per cent above the SPI projection for the same year (£80.0 billion: the SPI 2023–24 outturn projected to 2026 by PolicyEngine's data pipeline).[2] Three scaled variants, each over 20 draws paired with the unscaled central run, scale the variable in both the baseline and the shocked runs for people above the threshold. Scaling dividends by one factor (0.863) to the SPI total raises the central cost from £7.2 billion to £9.3 billion (a paired increase of £2.1 billion) and changes the Gini response by −0.11 points; scaling dividends band by band to the SPI band totals (top-band factor 0.52) gives £9.3 billion (an increase of £2.0 billion); and scaling savings interest by 0.761 to the SPI total of £20.4 billion gives £8.2 billion (an increase of £0.9 billion). These are sensitivities, not alternative headlines: the data do not show whether the gap reflects top-income dividends the projection misses or calibration error, and the draw spread does not capture it, because the capital uplift is the same in every draw.

Exposure index. Table 8 reruns the central scenario under five exposure indices: the C-AIOE, the unadjusted Felten et al. (2021) AIOE, the Eloundou et al. (2024) GPT-exposure beta, and the all-AI and large-language-model indices of Department for Education (2023). The quota is 7 per cent under every index, but a single draw realises a different number of displaced workers under each, and across the 50 exposure-proportional draws each additional million displaced adds £11.0 billion to the cost. I therefore shift each seed-0 result to the 50-draw mean displaced count along the slope estimated from those draws. Adjusted, the cost ranges from £4.3 billion to £8.2 billion; the Felten AIOE, which costs more than the C-AIOE at seed 0 only because its draw displaces more workers, costs £8.0 billion against £8.2 billion. The Eloundou index, which loads less displacement on the highest-paid groups, costs least. The Gini rises, and the transition share rises from the bottom to the top decile, under every index.

Table 8: Central scenario under five exposure indices (seed 0, adjusted to the mean displaced count)

Index	Displaced (m)	Cost (£bn) Raw	Cost (£bn) Adjusted	ΔPov. BHC adj. (pp)	ΔGini adj. (pp)	D1 (%)	D10 (%)
C-AIOE (central)	1.96	7.4	8.2	+2.01	+1.54	0.48	5.61
Felten AIOE	2.16	9.3	8.0	+2.84	+1.85	0.69	6.18
Eloundou beta	1.95	3.4	4.3	+2.16	+1.86	0.84	4.95
DfE AIOE	1.89	5.9	7.5	+2.37	+1.79	0.73	4.96
DfE LLM	1.99	5.1	5.6	+2.11	+1.67	0.67	6.24

Note: Single draw, seed 0, 2026. Adjusted columns subtract the 50-draw OLS slope of each outcome on the displaced count (exposure-proportional draws) times the draw's deviation from the 50-draw mean count. D1 and D10: percentages of persons in the bottom and top baseline deciles who move to unemployment.

[2]The projection is `incomes_projection.csv` in the policyengine-uk-data repository (commit fa4ebe9), the source of Microcosm's own SPI calibration targets. Microcosm matches the SPI targets band by band up to £1 million of total income; in the untargeted band above it Microcosm holds £23.96 billion of dividends against £12.37 billion in the projection.

4.6 Where the displaced wage bill goes

The central scenario imposes the capital uplift independently of the labour shock, and its government balance counts no corporation tax. Table 9 replaces the uplift with an explicit account of where the displaced wage bill goes, on the seed-0 draw where every flow is available. Displaced workers lose £97.1 billion of earnings and survivors gain £30.9 billion, so labour income falls by £66.2 billion. The central scenario instead adds £48.0 billion of household capital income (£37.0 billion of dividends and £11.1 billion of interest).

In the closed account a share ϕ of the net labour-income fall reappears as company profit, which pays corporation tax at 25 per cent; companies pay out 50 per cent of post-tax profit, and a share s of that reaches UK households, the remainder going to overseas owners and pension funds, whose returns the household simulation does not tax. Household dividends pay tax at the 35 per cent average rate the simulated capital channel implies. The interest uplift has no counterpart in this account. The closed balance keeps the displacement and survivor-wage channels and their interaction and replaces the capital channel with corporation tax and dividend tax on the implied payout.

Table 9: A closed account of the displaced wage bill (seed 0, £bn a year)

ϕ	UK share s	Profits	Corporation tax	Household dividends	Dividend tax	Exchequer cost
0.5	0.5	33.1	8.3	6.2	2.2	13.8
0.5	1.0	33.1	8.3	12.4	4.4	11.6
1.0	0.5	66.2	16.6	12.4	4.4	3.3
1.0	1.0	66.2	16.6	24.8	8.8	−1.1

Note: Profits are ϕ times the net labour-income fall (£66.2 billion). Payout ratio 50 per cent. The seed-0 central cost with the exogenous capital uplift is £7.4 billion. Static accounting: no corporation-tax incidence, allowances or profit shifting, and dividend tax at the capital channel's average rate. Adding the £8.5 billion of net employer NI saved to profits would raise each profit figure by ϕ times that amount.

The closed account returns less than the central cost only when nearly all of the net labour-income fall becomes profit ($\phi = 1$); at $\phi = 0.5$ the cost is higher than the central cost whatever the UK share. The central capital uplift adds more dividend income (£37.0 billion) than the closed account delivers to households even at $\phi = 1$ and $s = 1$ (£24.8 billion), so the central scenario's sign rests on a capital uplift that the labour shock alone does not finance.

Table 10 replaces the static dividend step with simulated household dividends. Each row reruns the central displacement and survivor-wage draw over 20 seeds, paired person by person with the central run, and changes only capital income: proportional uplifts of interest and dividends of 10, 20 and 60 per cent, the central uplift applied to dividends alone, and closed variants in which household dividends rise, pro rata to baseline holdings, by the post-tax profit $(1-0.25)\,\phi\,(W-S)$ times the payout share, where W is the draw's displaced wage bill and S the survivors' gain. The simulated balance has no corporation tax, so the cost of each closed row deducts corporation tax at 25 per cent on $\phi(W-S)$, as Table 9 does; the closed rows therefore replace that table's dividend-tax step, computed at an average rate, with simulated household dividends.

Table 10: Capital-channel variants on the central draw (20 paired draws)

Capital variant	Exchequer cost (£bn/yr)	Corporation tax (£bn)	Δ Poverty BHC (pp)	Δ Gini (pp)	Capital income added (£bn)
pending					

Note: Means ± standard errors of the mean over seeds 0–19; every row shares the central displaced set and survivor-wage uplift. ϕ is the share of the net labour-income fall that becomes profit; payout is the share of post-tax profit paid to UK households. The cost of a closed row is net of the corporation tax in the next column.

4.7 A benchmark against the OBR

The OBR's March 2026 outlook includes a technological-displacement scenario in which equilibrium unemployment rises to 5.5 per cent, against 4.1 per cent in the central forecast for 2030; higher productivity leaves GDP unchanged and real earnings do not rise, so the labour share falls and the profit share rises (OBR, 2026a, Box 2.2). Borrowing is higher than in the central forecast by an average of £9 billion a year: receipts are £6 billion lower, from lower personal tax and VAT net of higher corporation tax, welfare spending is £3 billion higher, and debt interest rises with the extra borrowing (OBR, 2026a, para. 6.18).

I simulate the closest equivalent the household model allows. Displacement adds 1.4 points of Microcosm's 34.6 million labour force to unemployment, which is 1.67 per cent of employees under exposure-proportional incidence, with no survivor-wage change and no capital shock (20 draws). Table 11 sets the simulated change in the government balance beside the OBR's lines. The simulation costs £9.7 billion, about £20,200 per displaced worker. Its receipts line, £9.3 billion, exceeds the OBR's £6 billion, and its welfare line, £0.7 billion, falls short of the OBR's £3 billion.

The two sets of numbers measure different things in four ways. First, timing: the OBR figures average the forecast years, over which the unemployment gap builds up as the central forecast falls back to its equilibrium rate by 2030, whereas the simulation applies the full 1.4-point rise in one year, so the OBR's average is smaller than its own effect once the full rise is in place. Second, coverage: the OBR's receipts include VAT on lower consumption and corporation tax on higher profits, and its borrowing includes debt interest; the simulation has neither consumption nor profit responses. Third, the OBR's welfare figure covers all welfare spending, including contributory benefits that the simulation does not pay to the newly displaced. Fourth, the OBR's scenario does not allocate unemployment across occupations, whereas exposure-proportional incidence displaces higher earners, whose income tax and NI make up most of the simulated cost. The simulated balance also excludes the public-sector pay bill of the displaced, £5.8 billion. The long-term counterpart in the OBR's fiscal risks report, in which the wage share of GDP halves by 2075–76, lowers tax receipts by nearly 4 percentage points of GDP because wages and salaries carry an average effective tax rate of 46 per cent against 27 per cent on profits (OBR, 2026b, Box 4.1).

5 Policy responses

The existing system already absorbs part of the shock (Section 4.4). This section prices three stylised additions inside the shocked world. Each is simulated on top of the central scenario for

Table 11: The OBR technological-displacement scenario and a simulation of the same unemployment rise (£bn a year)

	This paper (2026, 20 draws)	OBR (average over the forecast)
Income tax	5.2	
Employee NI	1.0	
Employer NI	3.1	
Receipts lost	9.3	6
Universal Credit	0.67	
Other cash benefits	0.04	
Welfare added	0.71	3
Childcare support	−0.23	–
Statutory sick, maternity and paternity pay	−0.06	–
Debt interest	–	included
Change in borrowing	9.7	9
Memo: household basis (income tax, employee NI, benefits)	6.6	–
Memo: public-sector pay bill of the displaced	5.8	–

Note: Positive values add to borrowing. Simulation: 1.67 per cent of employees displaced, exposure-proportional, no survivor-wage or capital change; means over 20 draws. Childcare support and statutory pay are in the model's benefit total but are not welfare in the OBR's sense, so they are separate lines. The OBR's receipts include VAT and corporation tax, which the simulation holds fixed or omits, and its borrowing includes debt interest.

2026 only, and the estimand is the shocked-plus-reform world minus the shocked world, with poverty lines held at their shocked values.

R1 is a wage-insurance scheme in the spirit of trade-adjustment proposals: displaced workers receive 50 per cent of lost earnings, capped at £15,000 a year. I implement it as a post-simulation transfer that is untaxed and disregarded by means tests, so its gross cost equals its net cost and sits at the top of the range of possible designs. R2 raises the UC standard allowance by 20 per cent, and R3 suspends the benefit cap and cuts the UC taper from 55 to 45 per cent; both are PolicyEngine parameter reforms, so their costs are net of withdrawal elsewhere in the system. The fiscal and distributional aims of these designs differ, as do their administrative requirements; Table 12 compares them on cost and on poverty only.

Table 12: Policy reforms on top of the central shock, 2026 (50 draws)

Reform	Annual cost (£bn)	Δ Poverty BHC (pp)	Δ Gini (pp)	Cost per BHC point (£bn)
R1 Wage insurance	26.7 ± 1.1	−1.89 ± 0.17	−0.98	14.1
R2 UC allowance +20%	6.9 ± 0.1	−0.64 ± 0.04	−0.32	10.8
R3 Cap suspension + taper cut	4.9 ± 0.1	−0.62 ± 0.04	−0.25	7.8

Note: Shocked-plus-reform minus shocked, exposure-proportional incidence, means ± standard deviations across 50 draws. Cost per point is the mean cost divided by the mean poverty reduction. R1 is untaxed and means-test-disregarded.

Wage insurance reduces BHC poverty by 2.9 times as much as the UC allowance increase at

3.9 times the cost, so it costs £14.1 billion per point of poverty reduced, against £10.8 billion for R2 and £7.8 billion for R3. Wage insurance pays in proportion to lost earnings, and the exposure-proportional shock displaces workers mostly in the upper half of the distribution: at seed 0, 63 per cent of R1 spending reaches baseline deciles 8–10, while R2 and R3 place 77 and 72 per cent of their spending in deciles 1–5. Under junior-concentrated and uniform incidence R1 costs £26.6 billion and £26.0 billion (50 draws). In the 50 per cent earnings-retention run a duration-consistent R1 costs £31 billion per poverty point against £11.5 billion for R2 (5 draws). Figure 5 shows the decile distribution of each reform's gains.

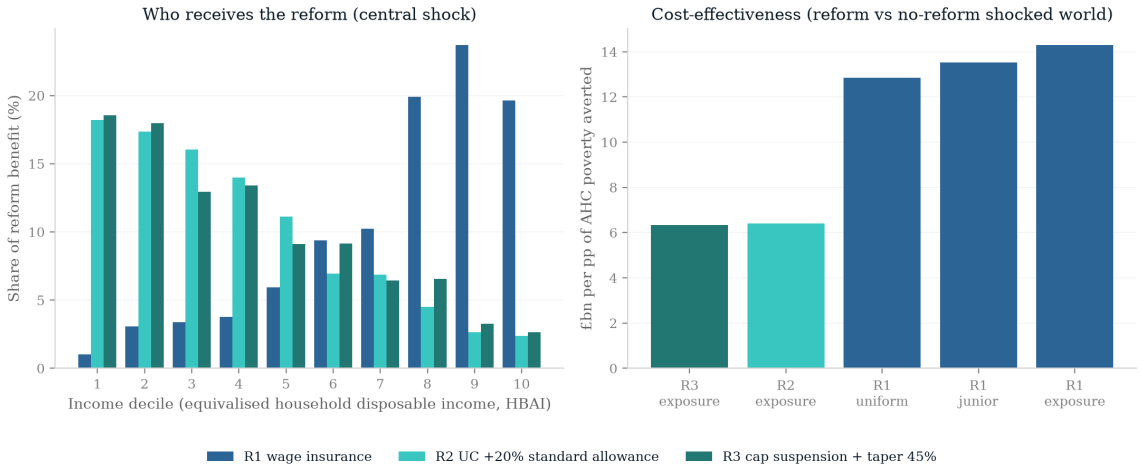


Figure 5: The three policy reforms on top of the central shock: fiscal cost, poverty and Gini effects, and the decile distribution of household gains. Shocked-plus-reform minus shocked, seed 0, 2026.

6 Conclusions

The results are scenario analysis. I do not forecast the pace or extent of AI adoption; I trace what the UK tax-benefit system would do if labour-market shocks of a given size, incidence and margin occurred under current rules. The observational record does not yet discriminate between the incidence views: the earliest evidence ranges from minimal effects on earnings and hours (Humlum and Vestergaard, 2025) to relative employment declines for young workers in exposed occupations (Brynjolfsson et al., 2026), none at the scale of the central shock.

Three findings do not depend on the central parameters. First, who loses the job moves the cost of a fixed head-count shock by £9.0 billion across the stylised rules, a difference that does not depend on the survivor-wage and capital receipts the rules share, because the Exchequer loses each displaced worker's income tax and NI and the UK tax base leans on the earnings of the occupations most exposed. Second, the margin changes how the shock is shared: the Exchequer absorbs more of each pound lost through wage cuts than through job loss, while job loss concentrates the household loss and raises absolute poverty by 1.95 points against 0.19 under wage cuts. Third, measured inequality rises under every displacement rule and grid cell examined, with or without the capital channel. The level and sign of the central cost, by contrast, turn on three assumptions with break-even values close to their central settings

(Table 7) and on how much of the displaced wage bill returns to UK households as taxable profit (Table 9).

The comparison with the Irish simulation of Doorley et al. (2026), which uses the same central conventions on SILC data and the SWITCH model, shows the same qualitative pattern (Table 13). The fiscal magnitudes are not comparable, because Doorley et al. (2026) report Exchequer effects as shares of a net revenue base that includes welfare spending.

Table 13: Comparison with the Irish simulation of Doorley et al. (2026)

	Ireland (SWITCH, SILC)	UK (this paper)
Central shock conventions	7% / +2.6% / +0.4pp	identical
Incidence of displacement	concentrated in upper deciles	0.80% (bottom) to 5.55% (top decile)
Gini response	rises in every scenario	rises in every family and grid cell

Note: Irish entries summarise the published report. The exercises use different microdata, occupational resolution and tax-benefit systems.

6.1 Limitations and further work

Exposure. Exposure varies across nine SOC2020 major groups through a chained crosswalk, a reconstructed complementarity index and ASHE weights covering 340 of 412 unit groups; the occupation of SPI-drawn top-income records is imputed. Quota rates run from zero (elementary occupations) to about 10 per cent (Table 1), so incidence is a between-group gradient. A UK-native index (Henseke et al., 2025) with detailed occupations imputed from the Labour Force Survey would remove both constraints.

Dynamics and behaviour. Displacement is a one-year stock with full-year zero earnings, no re-employment, no change in partners' hours, no redundancy pay and no running down of savings. UK evidence on earnings losses after displacement (Upward and Wright, 2019) would calibrate spell length and re-employment wages, and US evidence shows workers from AI-exposed occupations gaining from publicly funded retraining (Hyman et al., 2025), a margin the framework does not model. Contributory Jobseeker's Allowance is not paid to newly displaced workers, because the model carries only reported awards; and newly displaced households keep their baseline UC take-up flags. Reduced hiring of young workers, the margin on which the strongest early evidence lies (Brynjolfsson et al., 2026; Klein Teeselink, 2025; Hosseini Maasoum and Lichtinger, 2026a), is outside a stock-based framework.

Capital and the self-employed. The capital channel covers interest and dividends only, and its size depends on the dividend base (Section 4.5). The self-employed are outside the labour-market shocks.

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A Appendix

A.1 Seed and draw conventions

Displacement is a random draw of which employees lose their jobs, subject to occupation-group quotas. Within each draw, ordering keys are uniform within the group, so a record's inclusion probability does not depend on its calibrated weight. The results rest on the following bases.

- Means over 50 paired draws (seeds 0–49): the central scenario (Table 2) and its cost by component, the five displacement rules (Table 4), the policy reforms (Table 12), the decile transition and wage-gain profiles, and the regression slope used to adjust the index comparison.
- Means over 20 paired draws (seeds 0–19), with identical displacement in every run of a seed: the channel split and cost per displaced worker, the dividend-coverage runs, the capital variants (Table 10) and the capital break-even, the public-sector pay bill, the duration and take-up sensitivities, the 20-draw decile profile (Figure 2) and the gender displacement rates. The OBR benchmark (Tables 3 and 11) uses the same seeds at a lower displacement rate, so it pairs with the other runs by seed, not by person.
- Single draws at seed 0: the low and high presets, the shock-size grid and its variants, the wage-margin and mixed-margin tables, the survivor-wage panel and its break-even, the factor-income account, the absorption rates, the decile decomposition (Figure 3), the incidence figure, the index and ripple sensitivities and the caseloads. Seed 0 ranks 2 of 50 on the BHC poverty change and 19 on cost among the exposure-proportional draws, so its cost sits near the mean and its poverty change near the bottom of the range.
- Means over five draws (seeds 0–4): the 2008–09 backcast; 5 draws: the duration-consistent policy ranking.

A value written $a \pm b$ is a mean and the standard deviation across assignment draws; the standard error of the mean is $b/\sqrt{n}$ for n draws.

B Online appendix

This appendix is intended for online publication.

B.1 Wage and capital channel profiles

Survivor wage gains run from 2.55 per cent in the bottom decile to 2.73 per cent in the top (Figure 6), because complementarity varies little across the deciles in which people work. Capital income is concentrated: the top decile holds £73.8 billion of the £121 billion baseline total, the top two deciles 75.4 per cent and the bottom decile 2.4 per cent (Figure 7).

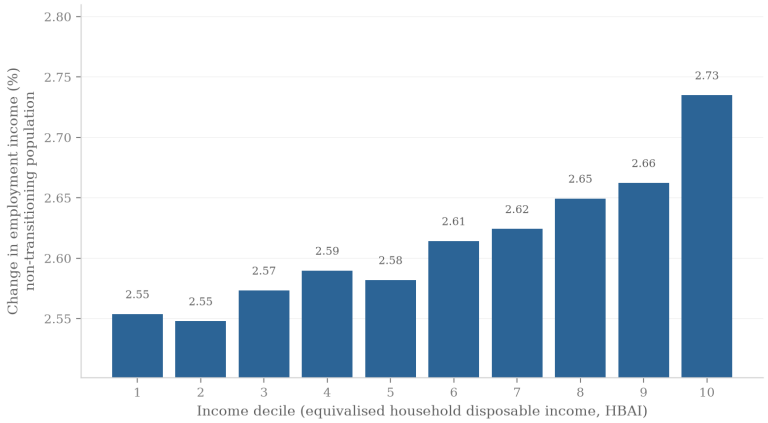


Figure 6: Percentage change in employment income among workers remaining in employment, by decile, central scenario. Means over 50 draws.

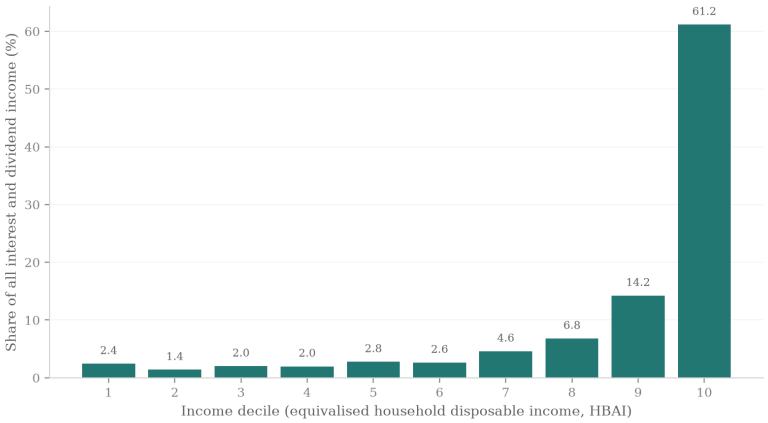


Figure 7: Capital-income channel by decile: a uniform proportional uplift, with pound gains concentrated where capital income is held.

B.2 Where the shock lands: baseline distributions

Figures 8–12 show the baseline household-level decile distributions of the income components through which the three channels operate (aggregate £billion per year by decile of equivalised household disposable income). Market income excluding capital (£1,540 billion in total) and income tax and National Insurance (£409 billion) are heavily concentrated in the top deciles—which is why displacement of high earners is fiscally expensive. Benefits (£332 billion) are spread across the lower and middle deciles, peaking in deciles 3 and 4. Capital income (£121 billion of interest and dividends) is the most concentrated series: the top decile holds 61.2 per cent of the total.

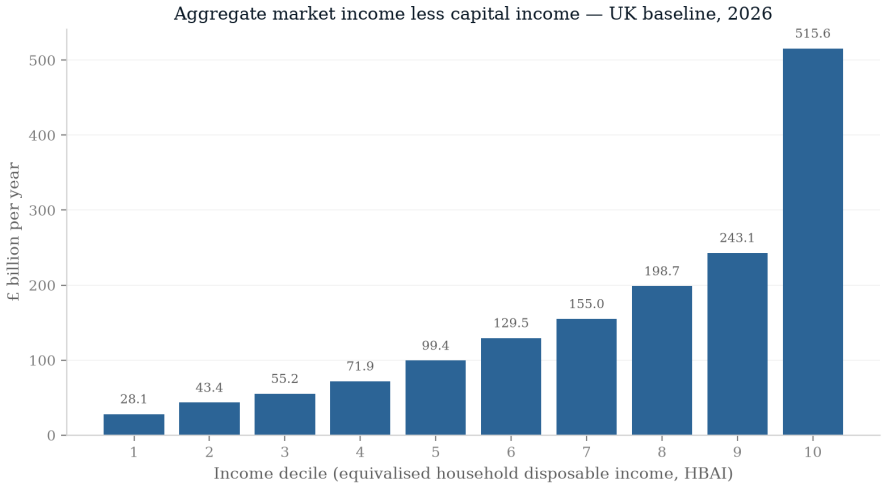


Figure 8: Baseline aggregate household market income (less capital income) by decile.

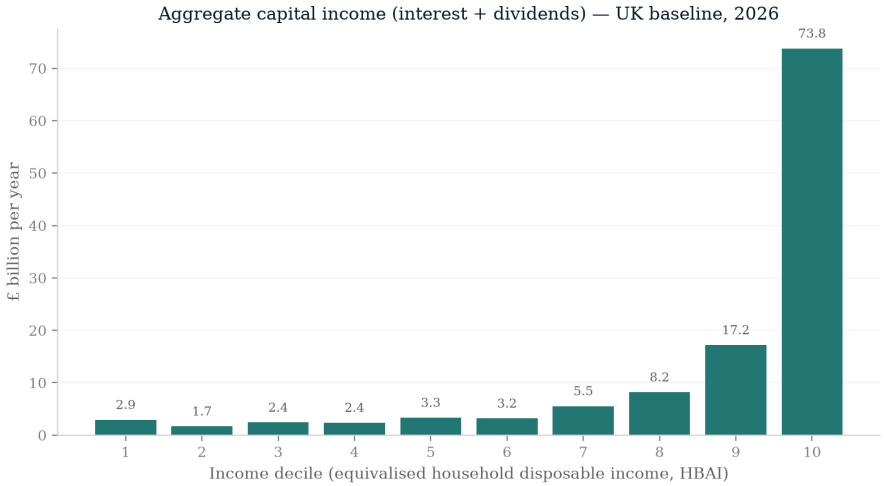


Figure 9: Baseline aggregate household capital income (interest and dividends) by decile.

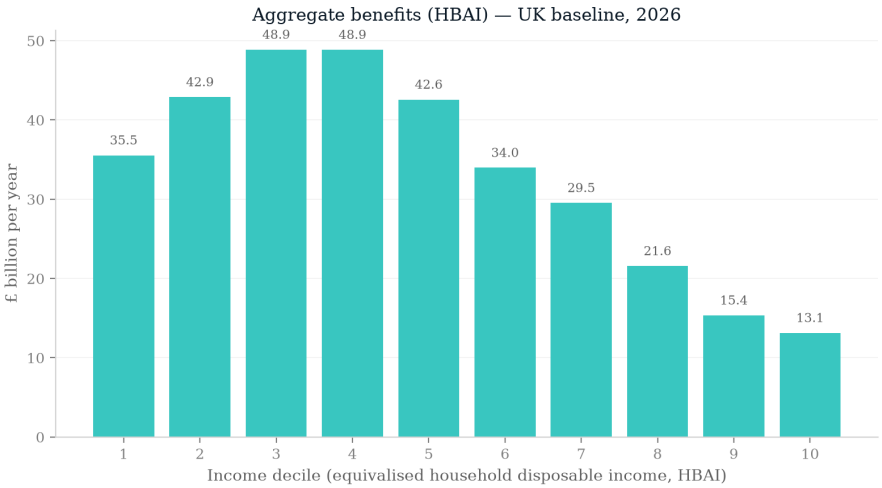


Figure 10: Baseline aggregate household benefits by decile.

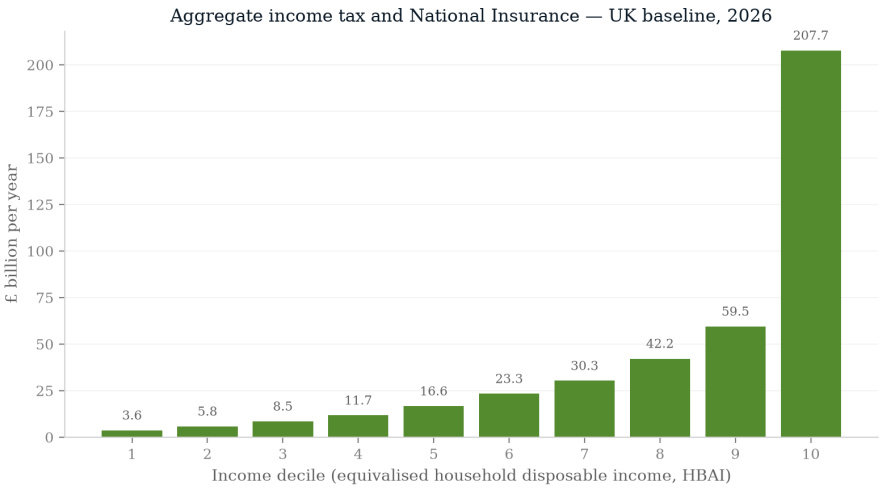


Figure 11: Baseline aggregate household income tax and National Insurance by decile.

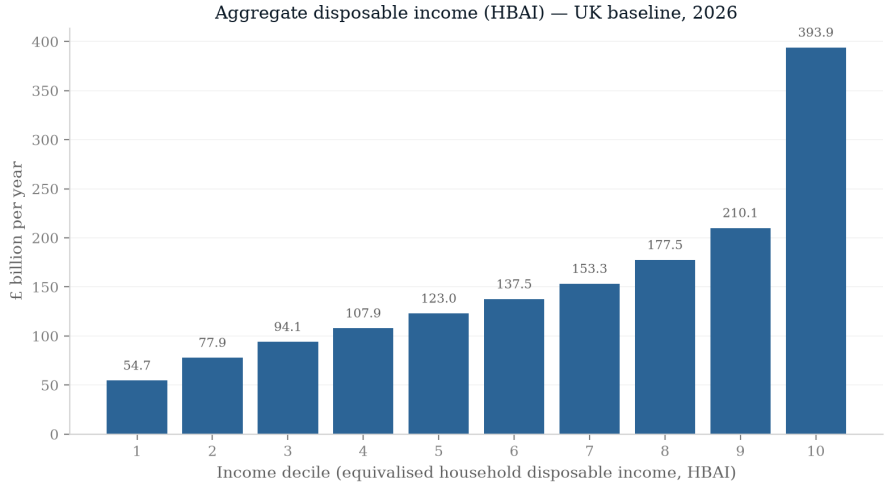


Figure 12: Baseline aggregate household disposable income (HBAI concept) by decile.

B.3 Exposure incidence before any simulation

Figures 13 and 14 show, without any shock, how workers in each income decile distribute across employment-weighted tertiles of AI exposure and of complementarity. The share of workers in the lowest exposure tertile falls with household income, and the lowest complementarity tertile shows a similar gradient—the descriptive facts that generate the simulated incidence in Section 4.

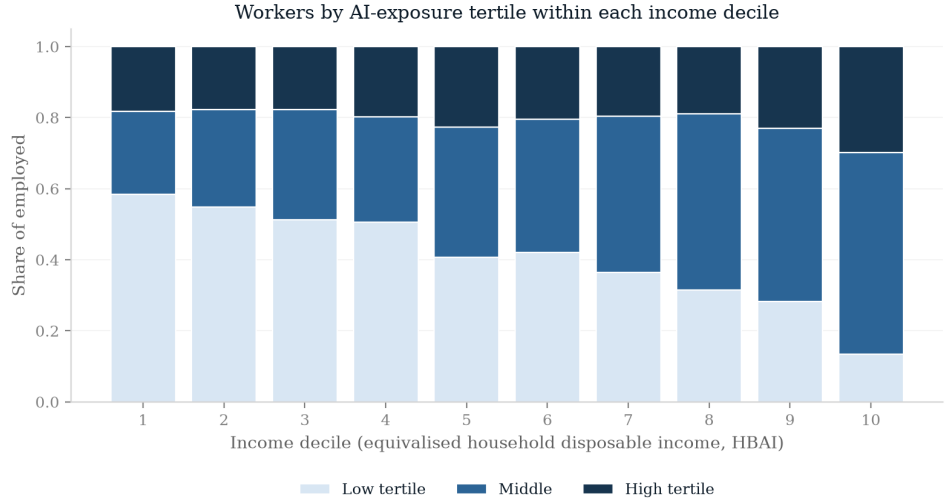


Figure 13: Distribution of employed persons across AI-exposure tertiles, by income decile.

B.4 Uniform-shock and alternative-index comparisons by decile

Figure 15 compares the central scenario against the same aggregate shocks allocated uniformly across occupations, decile by decile; Figure 16 repeats the central scenario with the Eloundou

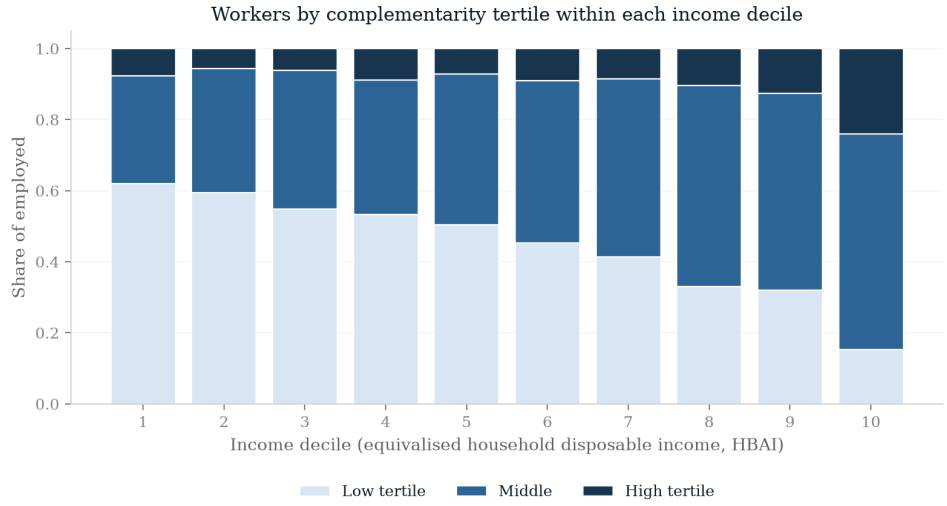


Figure 14: Distribution of employed persons across complementarity tertiles, by income decile.

et al. (2024) GPT-exposure index in place of the C-AIOE. The exposure-based allocation shifts losses towards deciles 8 and 9 relative to a uniform shock: in this single seed-0 draw, with each household weighted by its number of members (Figure 3 weights households once and gives smaller decile losses), they change by −4.4 and −4.8 per cent under exposure-based allocation against −2.7 and −1.5 per cent under the uniform shock, while the top decile changes by +0.2 and +0.7 per cent. Lower-decile comparisons are noisy at the single-draw level.

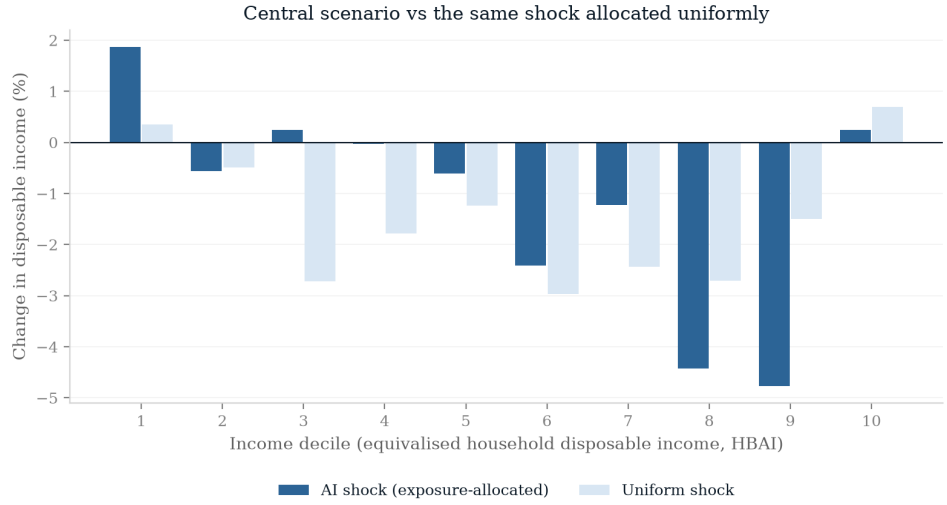


Figure 15: Disposable income change by decile: exposure-allocated versus uniform shock, central parameters (seed 0; each household weighted by its number of members).

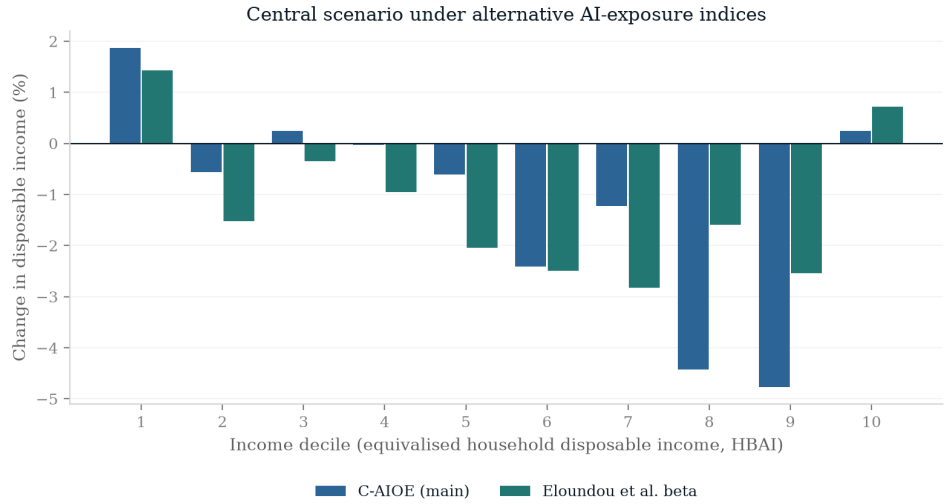


Figure 16: Disposable income change by decile under the C-AIOE and the Eloundou et al. exposure index (seed 0; each household weighted by its number of members).

B.5 The shock-size grid

To move beyond the presets I simulate a grid of 66 scenarios, crossing eleven displacement rates (0 to 10 per cent of employees, allocated in proportion to exposure, as in the presets) with six wage uplifts (0 to 5 per cent), each combined with the capital income channel, all under exposure-proportional incidence. Figure 17 maps the change in average household disposable income. The corners bracket the range: with a 5 per cent wage gain and no job loss, average disposable income changes by +4.6 per cent; with 10 per cent displacement and no wage gain it changes by −4.1 per cent. The capital channel shifts the whole surface upwards: on its own (no displacement, no wage gain) it raises average disposable income by +2.0 per cent. With no wage growth, average disposable income still rises at 3 per cent displacement (+0.48 per cent) and first falls at 4 per cent (−0.36 per cent); at the central 7 per cent displacement rate it falls with up to 4 per cent wage growth (−0.29 per cent) and rises slightly with 5 per cent (+0.17 per cent). Whether AI raises or lowers average living standards in this grid therefore depends on the balance between displacement and wage growth, and on the assumed capital-return shock, all of which are uncertain.

The grid's fiscal measure is narrower than the government balance behind the headline cost of Section 4.1: it is income tax plus National Insurance minus cash benefit expenditure as simulated on the household microdata, without employer National Insurance. Over the 50 central draws the narrow cash basis shows a net gain of £1.2 billion, against a cost of £7.4 billion on the government balance; the gap of £8.6 billion matches the employer National Insurance component of £8.6 billion, so the two measures should not be read off the same surface. On the narrow basis the grid spans a gain of £42.6 billion (5 per cent wage growth, no displacement) to a loss of £22.2 billion in the worst cell (10 per cent displacement, no wage growth). The capital shock alone adds £16.8 billion of net revenue. At the central displacement rate, net revenue falls by £11.2 billion with no wage offset and rises by £3.0 billion with a 3 per cent wage uplift.

Figure 18 maps the change in the Gini coefficient across the same grid. The Gini rises in

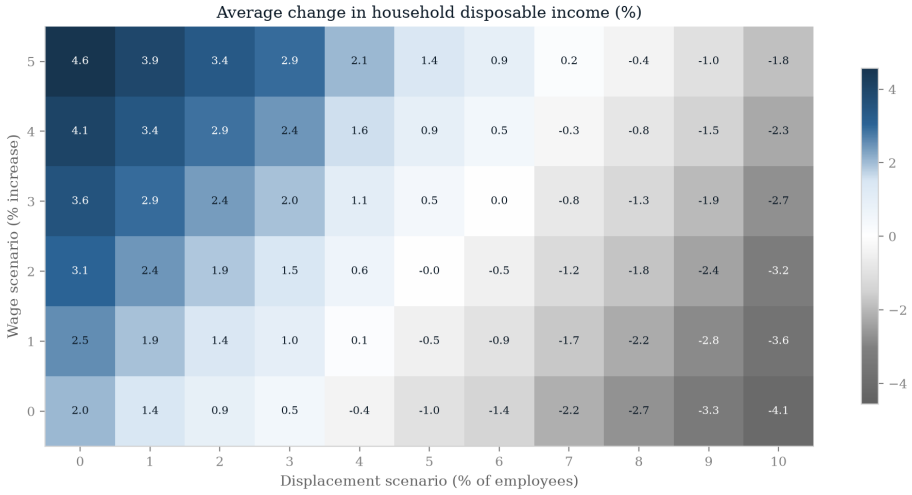


Figure 17: Change in average household disposable income (per cent) across the 66-cell grid of displacement rates and wage uplifts (single draw, seed 0).

all 66 cells, from +0.70 percentage points in the capital-only cell (no displacement, no wage gain) to +2.35 points at the far corner. The capital channel alone accounts for a substantial part of the rise, because the uplift accrues mainly to the top decile; displacement adds to it by pushing affected households down the distribution, and the wage uplift by lifting surviving workers further from workless and low-income households. Because each cell is a single draw on uneven weights, the Gini change is not strictly monotone in the displacement rate; the cross-draw spread reported for the central scenario (Section 4.1) indicates the scale of that noise. Within the model, inequality rises even though the displacement shock lands mainly on higher earners.

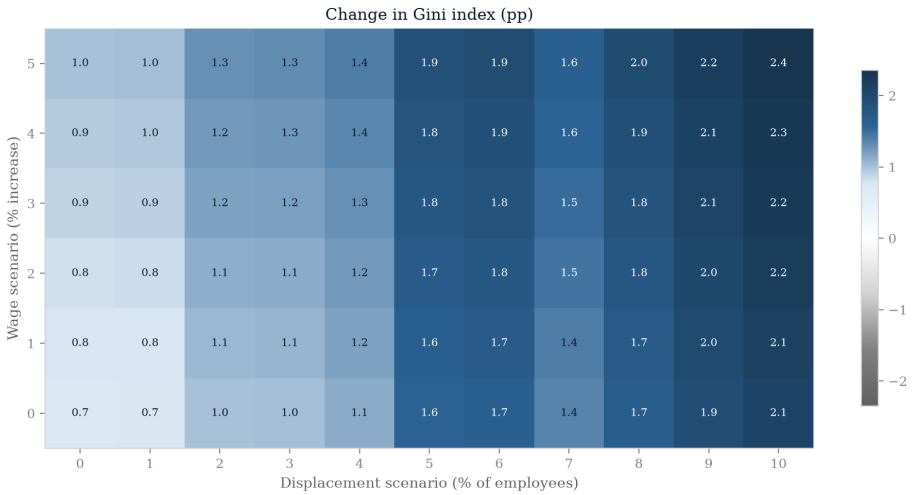


Figure 18: Change in the Gini coefficient of equivalised HBAI disposable income (percentage points) across the 66-cell scenario grid. The Gini rises in every cell.

Figure 19 facets the full 66-cell scenario grid by baseline income decile; cells report the change in mean household disposable income (HBAI concept), which across the grid as a whole ranges from +4.6 to −4.1 per cent. Losses deepen with displacement in every decile, though not monotonically in a single draw. In the cell with the largest loss (10 per cent displacement, no wage gain) the top decile's disposable income changes by −3.7 per cent; the bottom decile is subject to the ranking caveat of Section 4.2. Figure 20 removes the capital-return shock. Average disposable incomes fall relative to the capital-on grid, confirming that the capital channel raises average incomes while concentrating its gains at the top. Within this grid, without the capital shock the Gini of equivalised disposable income rises in 65 of the 66 cells, by up to 1.65 percentage points; the null $(0, 0)$ cell, which with the capital shock also off contains no shock at all, is exactly zero. Thus the positive Gini change in the shocked cells is not generated solely by the capital channel.

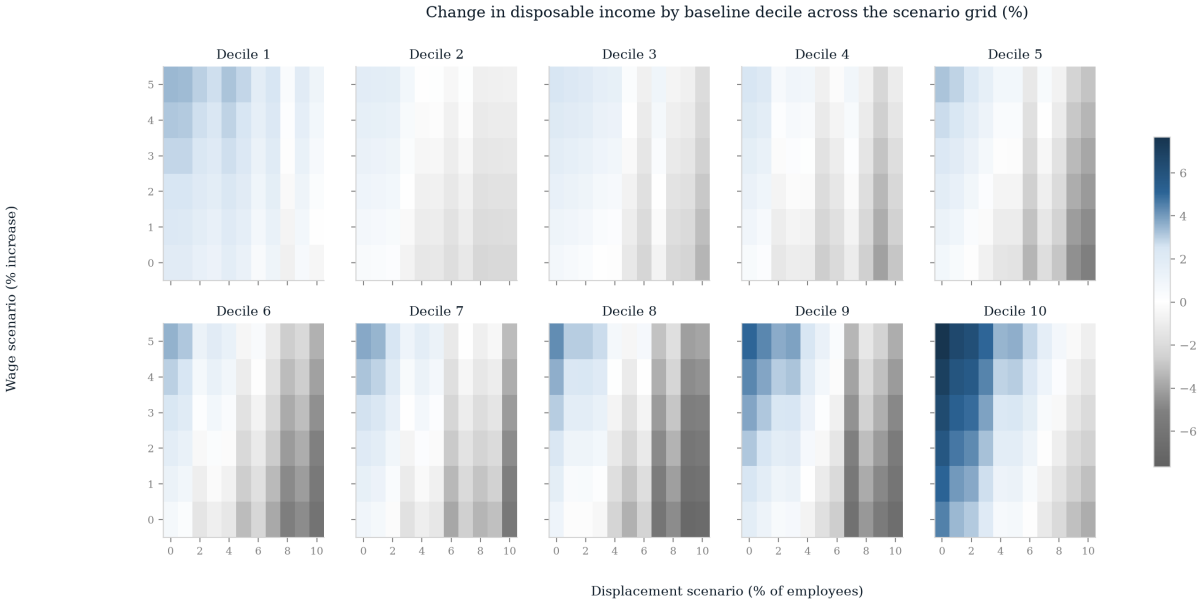


Figure 19: Change in household disposable income across the scenario grid, faceted by baseline income decile. Single draw, seed 0; each household weighted by its number of members.

B.6 Ripple-effect sensitivity

The displacement scenarios of the main text hold non-displaced wages at the uplifted baseline, so the ripple effects of Acemoglu and Restrepo (2022)—displaced workers bidding down the wages of the non-displaced—are absent. Here I price that channel in reduced form. After the central displacement draw, each displaced worker's labour supply is routed across destination SOC2020 major groups using a row-stochastic 9×9 routing matrix with a zero diagonal: the preferred source for this matrix, a pairwise retraining-cost matrix from the Hosseini Maasoum and Lichtinger (2026b) data release, does not exist (the release contains only occupation-level measures, with no origin–destination pairs), so I fall back on employment-proportional routing, with each origin row proportional to destination employment shares from the same ASHE 2025 employment weights used throughout the paper. For each destination group I compute the proportional labour-supply inflow s_l (routed displaced FTE over baseline employment), and

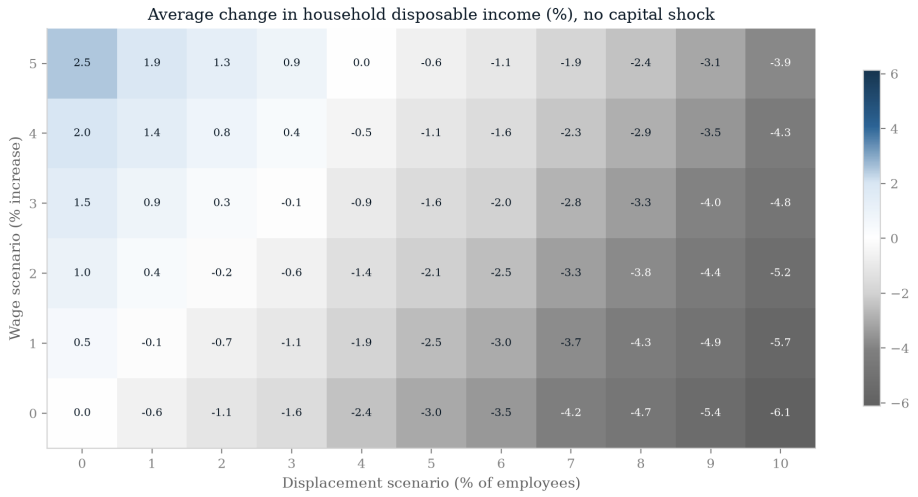


Figure 20: Average change in household disposable income across the scenario grid with the capital-return shock switched off. Single draw, seed 0.

every non-displaced employee in group l takes a wage cut of $\eta \times s_l$ of baseline earnings, composed additively with the +2.6 per cent uplift. The parameter η is an *assumed inverse-demand slope*—the proportional wage decline per unit proportional labour-supply inflow—not a sourced own-wage elasticity, and no conclusion about whether the central scenario is conservative should be drawn from any single value of it; with 7 per cent of workers displaced the inflow shares are around 7 per cent, so $\eta = 0.3$ implies wage cuts of roughly 2 per cent across the non-displaced workforce. This is a directional sensitivity, not an implementation of the Acemoglu and Restrepo (2022) propagation matrix: there is no task structure, no equilibrium feedback, and the routing is exogenous. Table 14 reports the results for $\eta \in \{0, 0.15, 0.3, 0.5\}$.

Table 14: Central scenario with reduced-form ripple effects

η	Cost (£bn)	ΔPov. BHC (pp)	ΔPov. AHC (pp)	ΔGini (pp)
0 (central)	7.4	+1.95	+2.15	+1.50
0.15	13.4	+2.03	+2.31	+1.46
0.30	19.5	+2.10	+2.56	+1.43
0.50	27.6	+2.26	+2.89	+1.38

Single draw, seed 0, 2026. η is the author-chosen wage sensitivity applied to each destination group's proportional labour-supply inflow. $\eta = 0$ reproduces the central scenario exactly.

Under this specification, the ripple channel loads mainly on the Exchequer. At $\eta = 0.3$ the fiscal cost rises by £12.2 billion, from £7.4 to £19.5 billion, because thin, widespread wage cuts across the non-displaced workforce forgo income tax and National Insurance at the margin. The distributional estimates move less: BHC poverty rises by 0.15 points more than in the central run (Table 14 gives the Gini change); the cuts are imposed on employed workers concentrated in the middle and upper deciles. Adding this particular ripple raises the simulated fiscal cost; other routing and wage-response assumptions could produce different results.